Time series Ex.1

1. Using the time-series data on retail sales ($Y_t$) and disposable income ($X_t$) of a country in Table 1 calculate the following:

(a) The *ex post* forecasts (2006-2010) for the naive, three-period moving average, trend, and causal models.

(b) The MAE, MAPE, MSE, and RMSE for the *ex post* period, 2006-2010.

(c) Is your choice of the "best" model for the *ex post* period different from the decision that was made when evaluating over the estimation period? Discuss.

**TABLE 1**

**Historical data for retail sales and disposable**

**personal income (billions of dollars) of a country**

| Year | RET_SALES | DIS_PY |
|---|---|---|
| 1990 | 375.2 | 710.0 |
| 1991 | 414.2 | 771.8 |
| 1992 | 458.5 | 834.8 |
| 1993 | 511.9 | 943.8 |
| 1994 | 542.0 | 1,033.7 |
| 1995 | 588.1 | 1,138.4 |
| 1996 | 656.4 | 1,248.8 |
| 1997 | 722.5 | 1,375.3 |
| 1998 | 804.2 | 1,546.5 |
| 1999 | 896.8 | 1,724.6 |
| 2000 | 957.4 | 1,914.3 |
| 2001 | 1,038.7 | 2,121.8 |
| 2002 | 1,069.3 | 2,255.1 |
| 2003 | 1,167.9 | 2,424.9 |
| 2004 | 1,281.7 | 2,662.1 |
| 2005 | 1,365.8 | 2,832.1 |
| 2006 | 1,435.9 | 3,007.6 |
| 2007 | 1,521.4 | 3,184.2 |
| 2008 | 1,629.2 | 3,467.9 |
| 2009 | 1,733.7 | 3,710.0 |
| 2010 | 1,807.2 | 3,949.1 |

2. Suppose that you work for an automobile association that publishes two-year forecasts for the price of leaded gasoline in its annual publication. You have collected the following information on the price of crude oil ( $X_t$ , dollars per barrel) ; and the price of leaded gasoline ($Y_t$ , dollars per gallon) in order to develop the best forecasting model.

| Year | Crude oil | Price of leaded regular |
|---|---|---|
| 1995 | 10.38 | 0.57 |
| 1996 | 10.89 | 0.59 |
| 1997 | 11.96 | 0.62 |
| 1998 | 19.46 | 0.63 |
| 1999 | 17.72 | 0.86 |
| 2000 | 28.07 | 1.19 |
| 2001 | 35.24 | l.31 |
| 2002 | 31.87 | 1.22 |
| 2003 | 28.99 | 1.16 |
| 2004 | 28.63 | 1.13 |
| 2005 | 26.75 | l.12 |
| 2006 | l 4.55 | 0.86 |
| 2007 | 17.90 | 0.90 |
| 2008 | 14.71 | 0.90 |

(a) Generate forecasts for the price of leaded gasoline using a naive model and a three-period simple moving average over the period 1998-2008.

(b) Suppose a trend model estimated over the period 1998-2008 yields the following result:

$$P_{leaded} = 1.022 + 0.001 * TIME$$

Generate forecasts for this model. Note that 1998-2008 is recoded as 1-11.

(c) Suppose a causal model was estimated over the period 1998-2008; the price of leaded gasoline is a function of the price of oil:

$$P_{leaded} = 0.444 + 0.025 * P_{crude}$$

(d) Calculate the MSE, RMSE, MAE, and MAPE for each of the models. Which model is the best? Why?

(e) **Forecast the** price of leaded gasoline for 2009 and 2010 using the naive model, the simple moving average model, and the trend model.

3. Use the following data on dividends per share (DPS) and earnings per share (EPS) for Weyerhaeuser Company:

| Year | DPS | EPS |
|---|---|---|
| 1995 | .53 | 1.01 |
| 1996 | .53 | 1.55 |
| 1997 | .53 | 1.53 |
| 1998 | .57 | 2.12 |
| 1999 | .72 | 2.68 |
| 2000 | .87 | 1.65 |
| 2001 | .87 | 1.11 |
| 2002 | .87 | 0.60 |
| 2003 | .87 | 0.91 |
| 2004 | .87 | 1.24 |
| 2005 | .87 | 0.88 |
| 2006 | .87 | 1.27 |
| 2007 | .90 | 2.12 |
| 2008 | 1.15 | 2.68 |
| 2009 | 1.20 | 2.83 |
| 2010 | 1.20 | 1.87 |

(a) Generate naive and three-period moving average forecasts for dividends paid per share over the estimation period 1998-2007.

(b) Suppose a trend model estimated over the period 1998-2007 yields the following result:

$$DPS_t = 0.694 + 0.024 * \text{time}$$

Generate the forecast values for dividends per share over the period 1998-2007.

(c) Suppose a causal model is estimated over the period 1998-2007; dividends per share are a function of earnings per share:

$$DPS_t = 0.961 - 0.091 * EPS.$$

Generate the forecast values for dividends per share over the period 1998-2007.

(d) Calcuiate the MAE, MAPE. MSE, and RMSE for the naive, the three-period moving average, the time trend, and the causal models. Which model is the best? By what measure?

(e) Generate the *ex post* forecasts for dividends per share for 2008, 2009, and 2010.

(f) Calculate the MAE, MAPE, MSE, and RMSE for the *ex post* period. Which model is the best? By what measure?

(g) What conclusions can you draw about model evaluation over the estimation period and its predictive accuracy over an *ex post* forecast period?

4. Suppose $Cov(X_t,X_{t-k}) = \gamma_k$ is free of t but that $E(X_t)=3t$.

(a) Is $\{X_t\}$ stationary?

(b) Let $Z_t= 7-3t+X_t$. Is $Z_t$ stationary?

5. Suppose $Y_t = A+ X_t$, where $\{X_t\}$ is stationary and A is random but independent of $\{X_t\}$. Find the mean and covariance function for $\{Y_t\}$ in terms of the mean and autocovariance function for $\{X_t\}$ and the mean and variance of A.

6. Let $Y_1 = \theta_0 + e_1$, and for t > 1 define $Y_t$ recursively by $Y_t = \theta_0 + Y_{t-1} + e_t$. Here $\theta_0$ is a constant and $\{e_t\}$ is a white noise sequence. The process $\{Y_t\}$ is called a random walk with drift.

(a) Show that $Y_t$ my be rewritten by $Y_t = t\theta_0 + e_t + e_{t-1} + ... + e_1$.

(b) Find the mean function for $Y_t$.

(c) Find autocovariance function for $Y_t$.